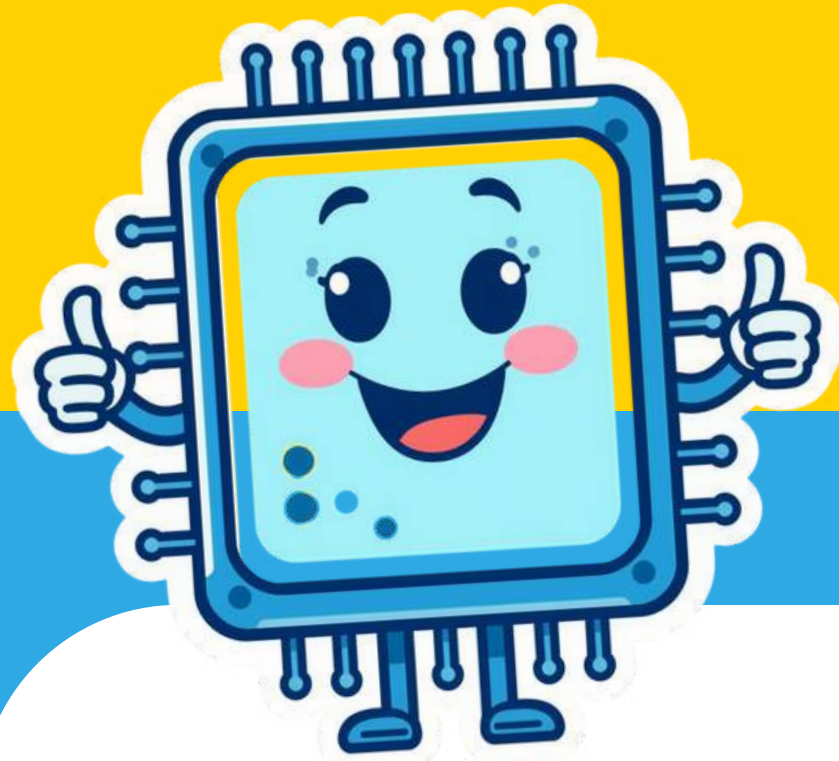




OP-AMP IC Audio Amplifier

Making Quiet Sounds Loud!



Listen Closely!

Can you hear this? Soft sound...
Now listen! LOUD sound!

Shhhh... whisper quietly. Now SHOUT!
Notice how sounds can be soft or loud?
Sometimes we need help making
sounds bigger!

Meet the Sound Superhero!

The Op-Amp IC is like a superhero that
takes tiny sounds and makes them LOUD
and clear for everyone to hear!



What is Sound?



Sound is vibration!

Sound travels through the air as tiny vibrations.



Microphone catches vibrations

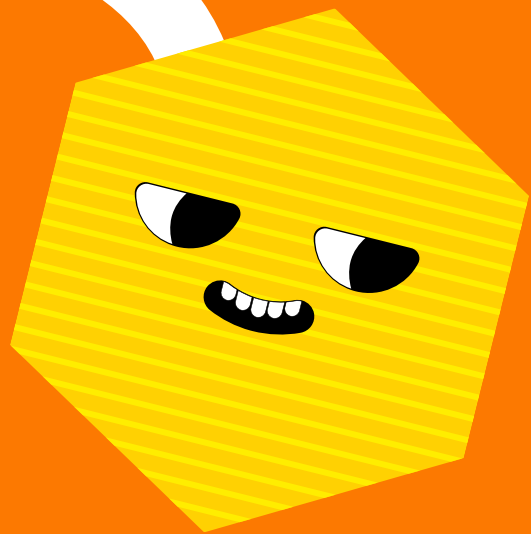
It picks up sound waves and captures them.



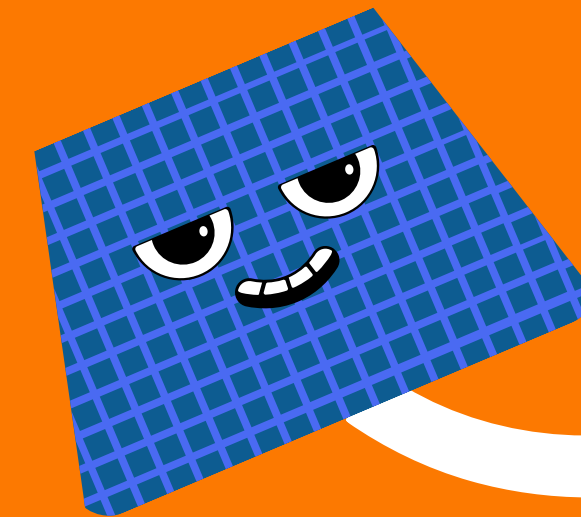
Changes into electrical signals

Vibrations become signals the amplifier can use!



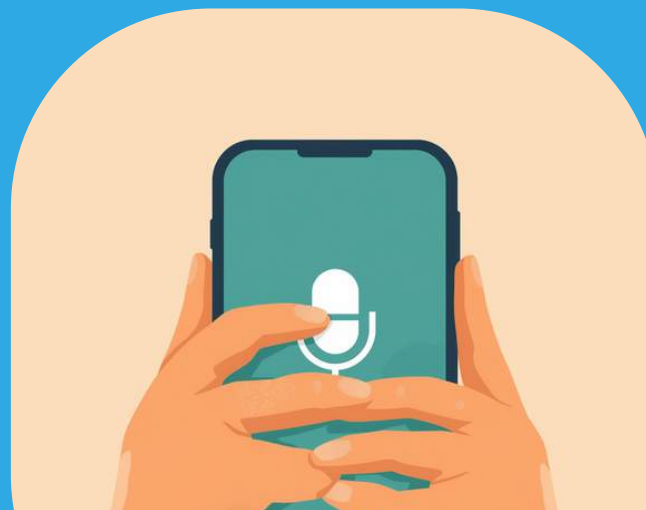


The Microphone Magic



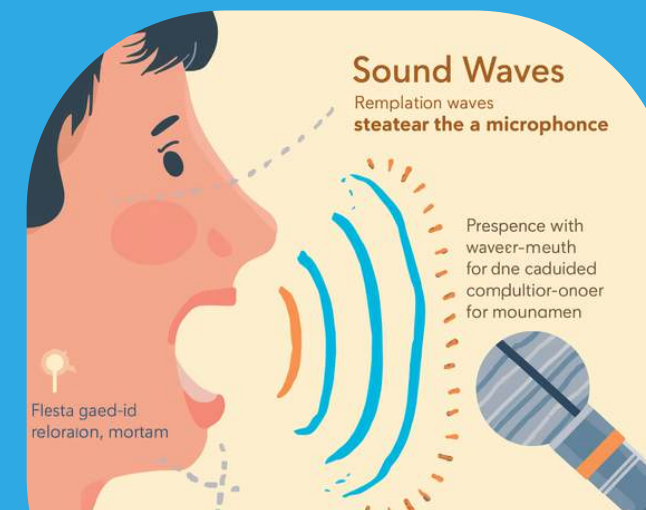
School
microphone

Teachers use microphones in assemblies to speak to everyone!



Phone
microphone

Your phone has a tiny microphone that catches your voice!



Sound waves
captured

Sound vibrations become electrical signals!



Signals to
amplifier

Microphones catch tiny sounds and turn them into signals for the amplifier!

What is an Amplifier?



Takes Small Input Sound



The microphone captures tiny sound signals and sends them to the amplifier.

Makes It Louder!



The amplifier boosts the small signal into a big, powerful sound wave.

Powered by Battery



The battery gives energy to the amplifier to do its magic work!

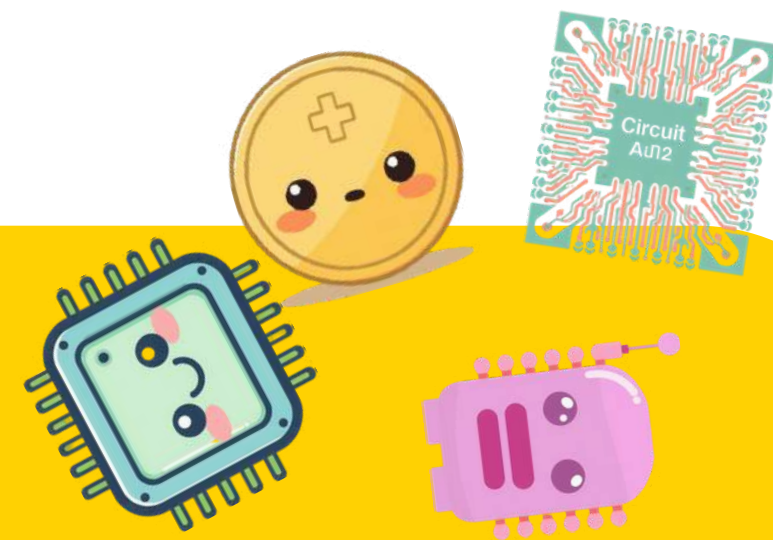
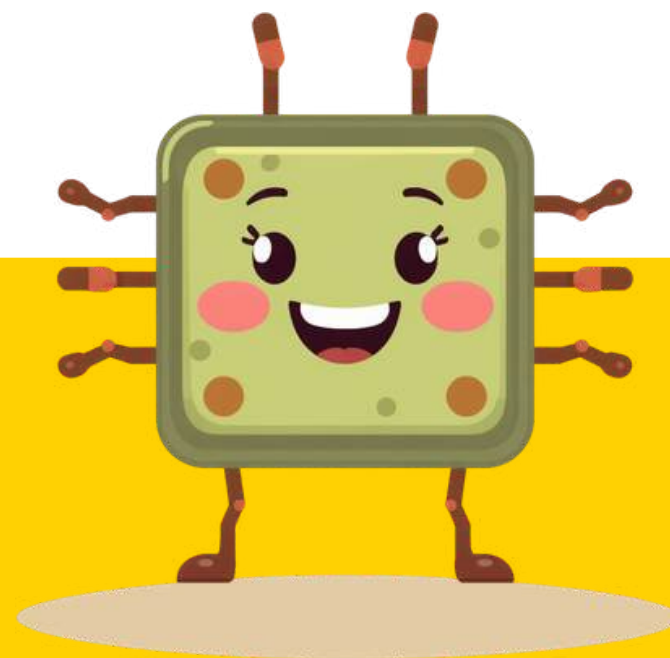
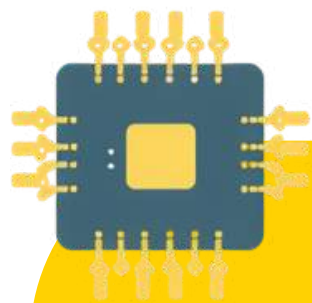
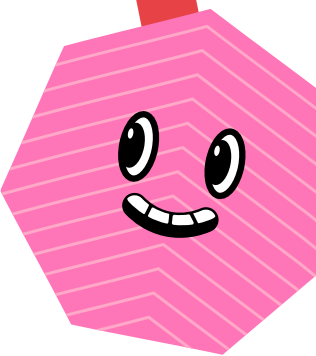
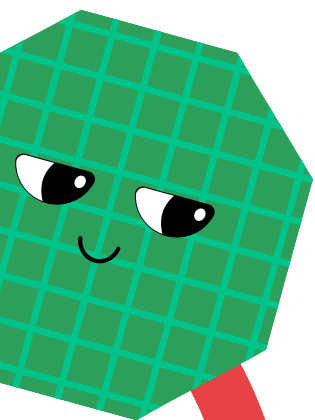
What is an IC?

IC = Integrated Circuit: A tiny chip with many parts inside

An IC chip is as small as your thumbnail or a coin, but it contains hundreds of tiny electronic parts working together!

Tiny but Powerful!

The Op-Amp IC has 8 little legs that connect it to other parts. Each leg has a special job to help amplify sound!



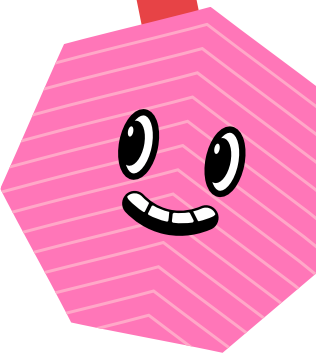
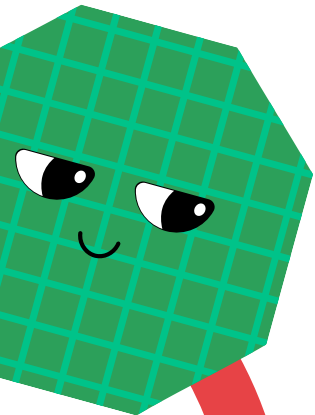
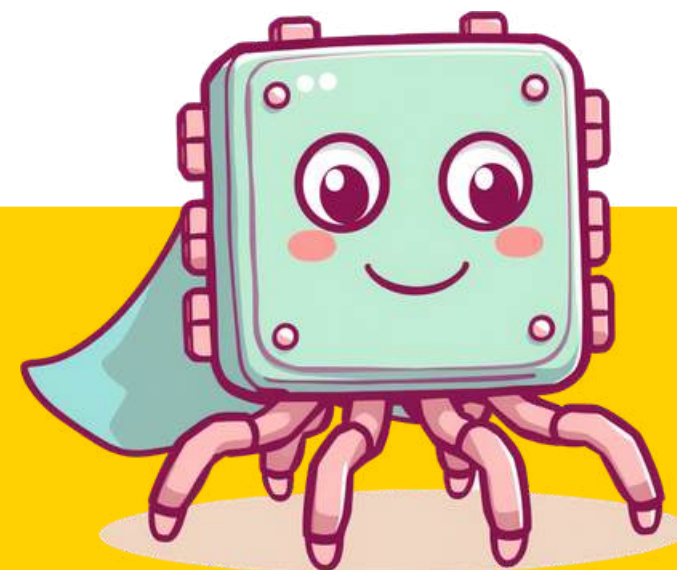
What is an Op-Amp?

Op-Amp = Special IC that amplifies sound signals

Op-Amp stands for Operational Amplifier. It's a special chip that takes tiny sound signals and makes them much bigger and stronger!

Uses battery power to make sounds louder

The Op-Amp uses energy from a battery to boost weak audio signals. It's like giving a small voice a megaphone to be heard clearly!



How It Works: The Sound Journey

SOUND →
MICROPHONE

Step 1

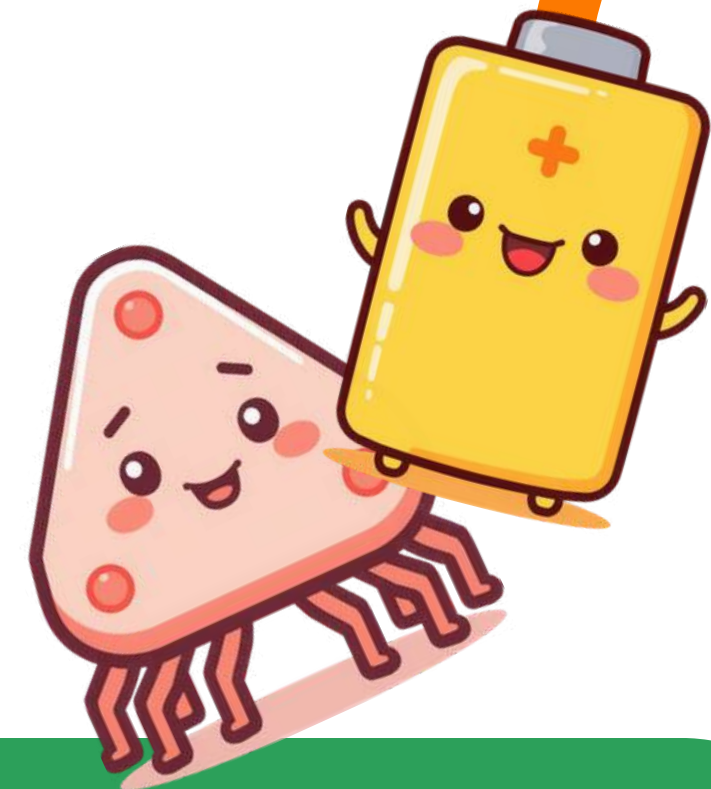
Tiny BLUE Signal →
OP-AMP IC

Step 2

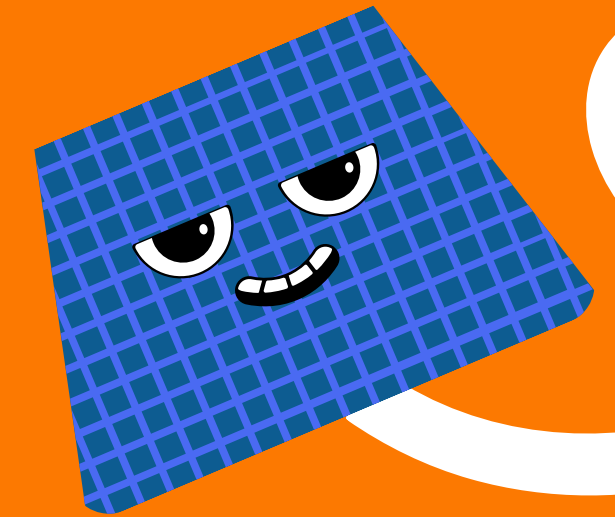
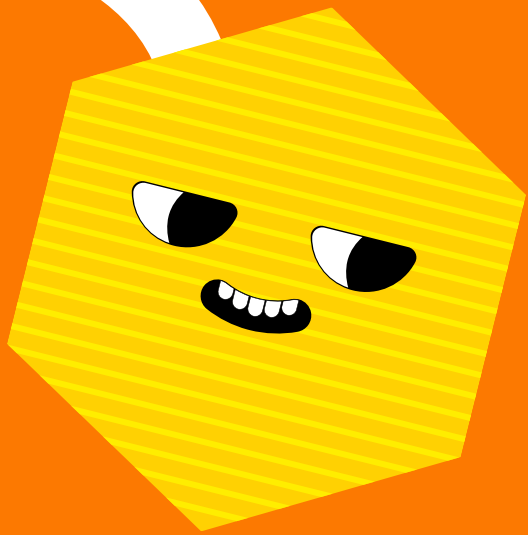
Big ORANGE
Output →
SPEAKER

Step 3

Yellow battery power feeds the
IC chip to make sounds LOUD!
Follow the colorful path!

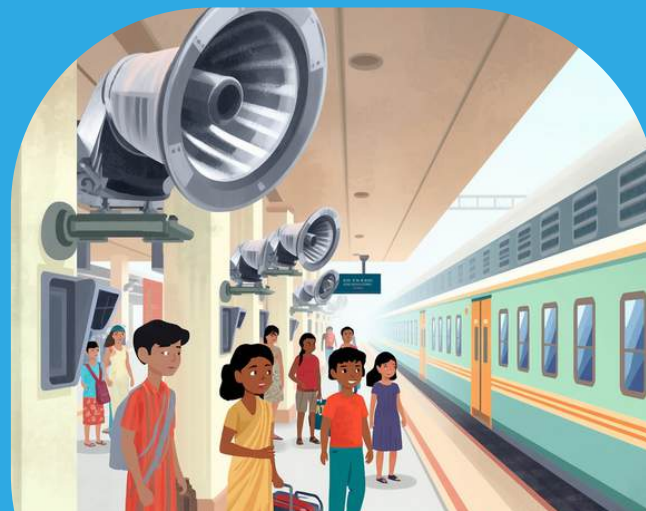


Amplifiers in Our Daily Life



School Assembly
PA System

Amplifiers help everyone hear in big school gatherings!



Railway Station
Announcements

At railway stations, amplifiers make sure you don't miss your train!



Mobile Phone
Speakers

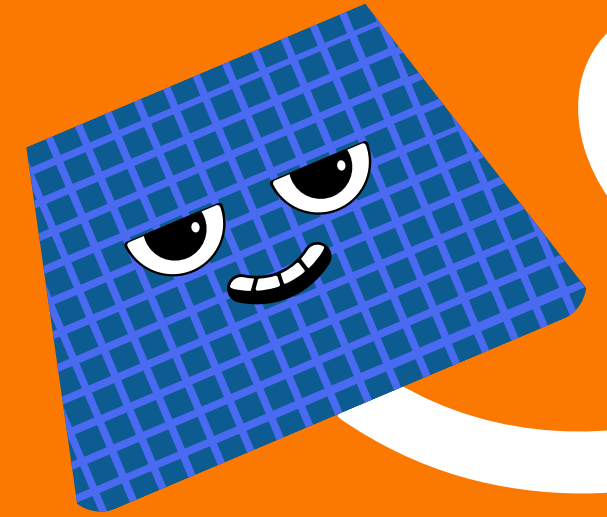
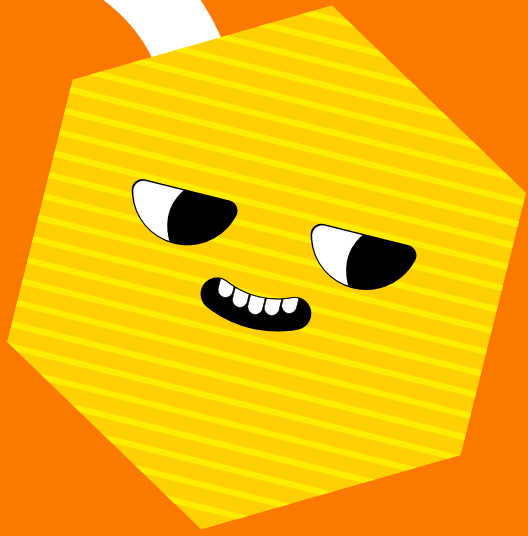
Your phone uses amplifiers so music sounds clear and loud!



Festival
Loudspeakers

Amplifiers help voices and music reach everyone during festivals!

Amplifiers in Action: India's Pride!



**Cricket Stadium
Speakers**

Amplifiers make the crowd roar and announcements loud in stadiums!



**ISRO Mission
Control**

Indian scientists use amplifiers to communicate with satellites in space!



**DRDO Research
Labs**

DRDO uses amplifiers in defense technology to keep India safe!



**Indian Scientists
at Work**

Amplifiers help scientists hear tiny signals from important equipment!

Recap: Vocabulary Time!

Sound, Signal, Amplifier, IC



Sound is vibration. Signal carries information. Amplifier makes sounds louder. IC is a tiny chip!

Op-Amp, Input, Output



Op-Amp amplifies signals. Input goes in, Output comes out bigger and louder!

Microphone, Loudspeaker, Battery, Audio



Microphone catches sound. Loudspeaker plays it. Battery gives power. Audio means sound!



Quiz Time!



What does an amplifier do?

It makes small sounds BIG and loud!



What does IC stand for?

Integrated Circuit – a tiny chip with many parts!



Name one place in India where amplifiers are used

Think of stadiums, railway stations, or festivals!





Thank You!

Keep exploring sounds around you!



Bonus: Fun Sound Experiment!



Clap Softly First

Start with a gentle clap. Notice how quiet it sounds?



Clap Loudly & Listen!

Now clap really loud! Can you hear the difference?

Imagine the Op-Amp Helping!

The Op-Amp IC makes small sounds BIG - just like your loud clap!

“ Try It Yourself - Explore Sounds! ”